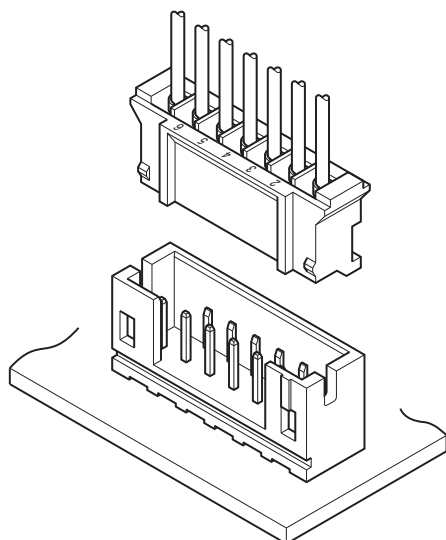


PH CONNECTOR

2.0 mm pitch/Wire-to-Board connectors/Crimp style and Mating style



With a mounting height of 8 mm and a width of only 4.5 mm in the top-entry version, this low-profile wire-to board connector features a pitch of 2.0 mm. The PH series is suitable for a wide range of applications, including use in high-density PCBs where space in electronic equipment is limited.

- Space-saving design
- High reliability connector
- Boxed-shaped shrouded header
- Excellent functionality in PCB mounting as a wire-to-board connector
- Available in surface mount style with SMT headers ranging from 2 to 16 positions.

Specifications

- Current rating: 2 A AC/DC (AWG #24)
- Voltage rating: 100 V AC/DC
- Temperature range: -40°C to +105°C
(including temperature rise in applying electrical current)
- Contact resistance: Initial value/ 10 mΩ max.
After test/ 20 mΩ max.
- Insulation resistance: 1,000 MΩ min.
- Withstanding voltage:
There shall be no breakdown or flashover while applying 800 VAC for one minute.
- Applicable wire range:
Conductor size/ AWG #32 to AWG #24
Insulation O.D./ ϕ 0.5 mm to ϕ 1.5 mm
- Applicable PC board thickness: 0.8 mm to 1.6 mm

* Please refer to the "Handling Precautions for Terminals and Connectors" on our website (listed in the "Technical Documents" column on the Product Information page) before use.

- * RoHS2 compliance
- * Dimensional unit: mm
- * Contact JST for details.

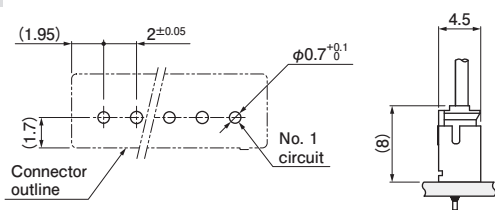
Standards

For information on overseas standard registrations, please refer to the "List of Registered Overseas Standards" on our website (listed in the "Technical Documents" column on the Product Information page).

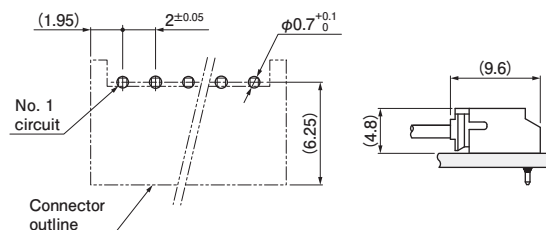
* Specifications registered to overseas standards may differ from the general specifications listed above.

PC board layout and Assembly layout (Through-hole type)

Top entry type



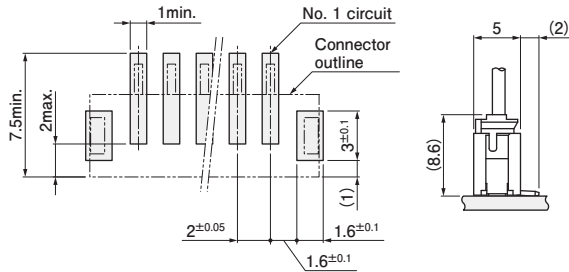
Side entry type



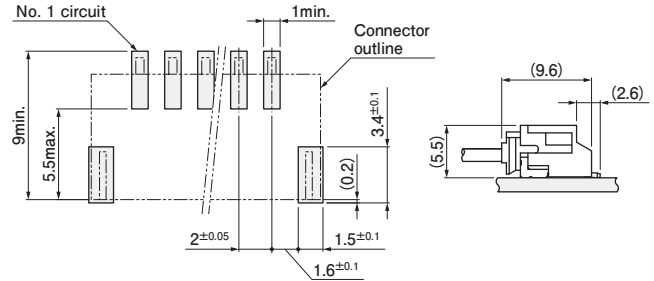
- Note: 1. The PC board layout figure shown is viewed from the connector mounting surface.
2. Tolerance for the PCB hole pitch shall be ± 0.05 and shall not accumulate.
3. Hole dimensions differ depending on the type of PCB and PCB drilling method.
When using PCB made of hard material composed of fiberglass cloth, please consider a larger hole diameter.
The above dimensions are for reference only. Please contact JST for further details.

PC board layout and Assembly layout (SMT type)

Top entry type

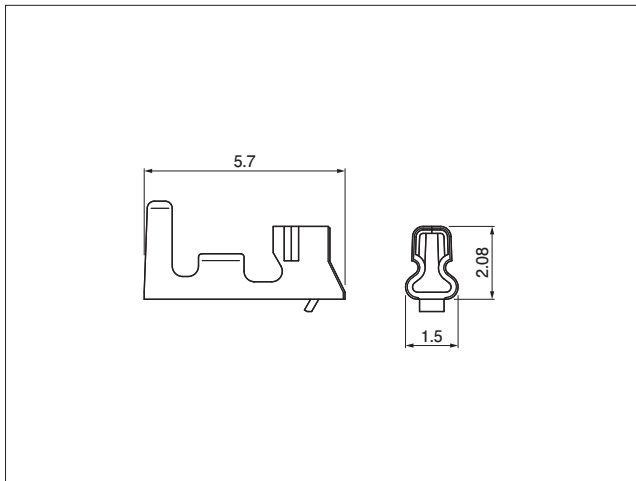


Side entry type



Note: 1. The PC board layout figure shown is viewed from the connector mounting surface.
2. Tolerance for the PCB pattern pitch shall be ± 0.05 and shall not accumulate.
The above dimensions are for reference only. Please contact JST for further details.

Contact



Model No.	Applicable wire range		Q'ty/ reel
	Conductor size AWG (mm ²)	Insulation O.D. (mm)	
SPH-004T-P0.5S	#32 to #28 (0.032 to 0.08)	0.5 to 0.9	10,000
SPH-002T-P0.5S	#30 to #24 (0.05 to 0.22)	0.8 to 1.5	8,000
SPH-002T-P0.5L	#28 to #24 (0.08 to 0.22)	0.8 to 1.5	

Material and Surface finish, etc.

Copper alloy, tin-plated

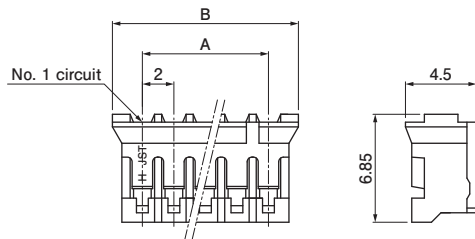
Note: 1. Contact JST for gold-plated products.
2. SPH-002T-P0.5L is a low-insertion force type contact for easier insertion/withdrawal operation. Please take the application and environment into consideration as the low-insertion force type contact is less resistant to vibration. Please note that the crimp height is different from the standard PH contact. Please contact JST for further details.

Crimping machine

Contact	Crimping machine	Applicator	Crimp applicator with dies
SPH-004T-P0.5S	AP-K2N	MKS-L-10	APLMK SPH004-05S
SPH-002T-P0.5S		MKS-L	APLMK SPH002-05S
SPH-002T-P0.5L			APLMK SPH002-05L

Note: Contact JST for fully automatic crimping applicator

Housing



No. of circuits	Model No.	Dimensions (mm)		Q'ty/bag
		A	B	
2	PHR-2	2.0	5.8	2,000
3	PHR-3	4.0	7.8	2,000
4	PHR-4	6.0	9.8	2,000
5	PHR-5	8.0	11.8	2,000
6	PHR-6	10.0	13.8	2,000
7	PHR-7	12.0	15.8	2,000
8	PHR-8	14.0	17.8	1,000
9	PHR-9	16.0	19.8	1,000
10	PHR-10	18.0	21.8	1,000
11	PHR-11	20.0	23.8	1,000
12	PHR-12	22.0	25.8	1,000
13	PHR-13	24.0	27.8	1,000
14	PHR-14	26.0	29.8	1,000
15	PHR-15	28.0	31.8	1,000
16	PHR-16	30.0	33.8	1,000

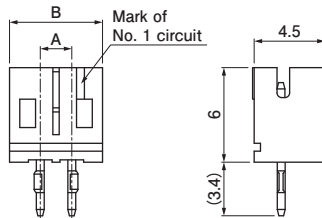
Material and Surface finish, etc.

PA, natural (white)

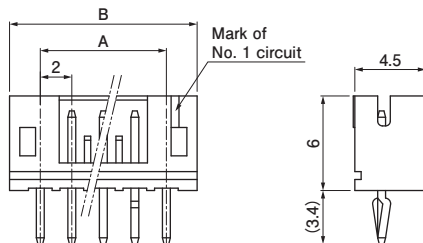
Note: For flame retardant grade of resin material used, please refer to the "List of Registered Overseas Standards" on our website (listed in the "Technical Documents" column on the Product Information page).

Header (Through-hole type)

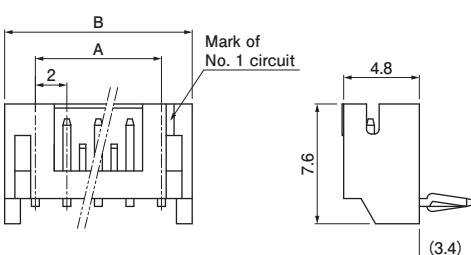
Top entry type <2 circuits>



<3 circuits or more>



Side entry type



No. of circuits	Model No.		Dimensions (mm)		Q'ty/box	
	Top entry type	Side entry type	A	B	Top entry type	Side entry type
2	B2B-PH-K-S	S2B-PH-K-S	2.0	5.9	1,000	1,000
3	B3B-PH-K-S	S3B-PH-K-S	4.0	7.9	1,000	1,000
4	B4B-PH-K-S	S4B-PH-K-S	6.0	9.9	1,000	500
5	B5B-PH-K-S	S5B-PH-K-S	8.0	11.9	1,000	500
6	B6B-PH-K-S	S6B-PH-K-S	10.0	13.9	1,000	500
7	B7B-PH-K-S	S7B-PH-K-S	12.0	15.9	500	500
8	B8B-PH-K-S	S8B-PH-K-S	14.0	17.9	500	250
9	B9B-PH-K-S	S9B-PH-K-S	16.0	19.9	500	250
10	B10B-PH-K-S	S10B-PH-K-S	18.0	21.9	500	250
11	B11B-PH-K-S	S11B-PH-K-S	20.0	23.9	500	250
12	B12B-PH-K-S	S12B-PH-K-S	22.0	25.9	400	250
13	B13B-PH-K-S	S13B-PH-K-S	24.0	27.9	250	250
14	B14B-PH-K-S	S14B-PH-K-S	26.0	29.9	250	250
15	B15B-PH-K-S	S15B-PH-K-S	28.0	31.9	250	200
16	B16B-PH-K-S	S16B-PH-K-S	30.0	33.9	250	200

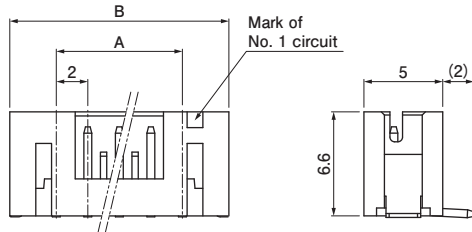
Material and Surface finish, etc.

Post: Copper alloy, tin-plated
Base housing: PA, natural (white)

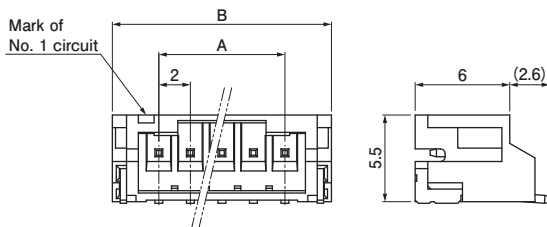
Note: 1. This product displays (LF) (SN) on a label.
2. For flame retardant grade of resin material used, please refer to the "List of Registered Overseas Standards" on our website (listed in the "Technical Documents" column on the Product Information page).

Header (SMT type)

Top entry type



Side entry type



Top entry type

No. of circuits	Model No.	Dimensions (mm)		Q'ty/reel
		A	B	
2	B2B-PH-SM4-TB	2.0	7.95	1,000
3	B3B-PH-SM4-TB	4.0	9.95	1,000
4	B4B-PH-SM4-TB	6.0	11.95	1,000
5	B5B-PH-SM4-TB	8.0	13.95	1,000
6	B6B-PH-SM4-TB	10.0	15.95	1,000
7	B7B-PH-SM4-TB	12.0	17.95	1,000
8	B8B-PH-SM4-TB	14.0	19.95	1,000
9	B9B-PH-SM4-TB	16.0	21.95	1,000
10	B10B-PH-SM4-TB	18.0	23.95	1,000
11	B11B-PH-SM4-TB	20.0	25.95	1,000
12	B12B-PH-SM4-TB	22.0	27.95	1,000
13	B13B-PH-SM4-TB	24.0	29.95	1,000
14	B14B-PH-SM4-TB	26.0	31.95	1,000
15	B15B-PH-SM4-TB	28.0	33.95	1,000
16	B16B-PH-SM4-TB	30.0	35.95	1,000

Material and Surface finish, etc.

Post: Copper alloy, tin-plated

Base housing: PA (Heat resistance), natural (ivory)

Reinforcement: Copper alloy, tin-plated

- Note: 1. This product displays (LF) (SN) on a label.
 2. Top-entry headers with suction tape are also available.
 3. For flame retardant grade of resin material used, please refer to the "List of Registered Overseas Standards" on our website (listed in the "Technical Documents" column on the Product Information page).

Side entry type

No. of circuits	Model No.	Dimensions (mm)		Q'ty/reel
		A	B	
2	S2B-PH-SM4-TB	2.0	7.9	1,000
3	S3B-PH-SM4-TB	4.0	9.9	1,000
4	S4B-PH-SM4-TB	6.0	11.9	1,000
5	S5B-PH-SM4-TB	8.0	13.9	1,000
6	S6B-PH-SM4-TB	10.0	15.9	1,000
7	S7B-PH-SM4-TB	12.0	17.9	1,000
8	S8B-PH-SM4-TB	14.0	19.9	1,000
9	S9B-PH-SM4-TB	16.0	21.9	1,000
10	S10B-PH-SM4-TB	18.0	23.9	1,000
11	S11B-PH-SM4-TB	20.0	25.9	1,000
12	S12B-PH-SM4-TB	22.0	27.9	1,000
13	S13B-PH-SM4-TB	24.0	29.9	1,000
14	S14B-PH-SM4-TB	26.0	31.9	1,000
15	S15B-PH-SM4-TB	28.0	33.9	1,000

Material and Surface finish, etc.

Post: Copper alloy, tin-plated

Base housing: PA (Heat resistance), natural (ivory)

Reinforcement: Copper alloy, tin-plated

- Note: 1. This product displays (LF) (SN) on a label.
 2. For flame retardant grade of resin material used, please refer to the "List of Registered Overseas Standards" on our website (listed in the "Technical Documents" column on the Product Information page).

Model number allocation

Contact

S PH - 002 T - P 0.5 S

Supply form: Strip form

Series name

Applicable wire range:
 Standard type 004...AWG #32 to AWG #28
 002...AWG #30 to AWG #24
 Low insertion force type
 002...AWG #28 to AWG #24

Surface finish: Tin-plated

Material: Copper alloy

Applicable post size

Type: S...Standard type, L...Low insertion force type

Housing

PH R - 2 -

Series name

Type: For receptacle contact

No. of circuits

Color: No indication...Natural (White),
 BK...Black, BL...Blue, D...Orange, DPK...Dark pink,
 FY...Fluorescent yellow, H...Gray, L...Lemon yellow,
 LE...Light blue, M...Green, N...Brown, P...Purple,
 PK...Pink, R...Red, TR...Tomato red, Y...Yellow

Header (Through-hole type)

B 2 B - PH - K - S

Header type: B...Top entry type
 S...Side entry type

No. of circuits

Assembly product

Series name

Clinched (Kinked) / Not clinched: Clinched

Color: S...Natural (White),
 No indication...Ivory, E...Blue, FY...Fluorescent yellow,
 H...Gray, K...Black, L...Lemon yellow, LE...Light blue,
 M...Green, N...Brown, O...Orange, P...Purple, PK...Pink,
 R...Red, TR...Tomato red, Y...Yellow

Header (SMT type)

B 2 B - PH - SM 4 - -TB

Header type:
 B...Top entry type
 S...Side entry type

No. of circuits

Assembly product

Series name

SMT type

Product version

Color: No indication...Natural (Ivory),
 DPK...Dark pink, E...Blue, K...Black,
 M...Green, L...Lemon yellow, R...Red

Packaging style: Embossed taping

Suction tape: No indication...No, T...Yes